

HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map

Flood mapping was derived from Sentinel-1 SAR (scene S1A_DEMO_PESHAWAR_20260613) rather than optical imagery because cloud cover reached 42%, exceeding the 30% threshold for reliable optical flood detection; SAR was selected for its all-weather, day-night surface-water sensitivity. The analysis applied a SAR VV/VH ratio index with a mean value of 0.24, consistent with extensive smooth open-water backscatter and saturated surfaces. Hazard model confidence is highest for flood (0.91), reinforcing the CRITICAL classification, while earthquake risk is rated MEDIUM (confidence 0.67) and landslide LOW (confidence 0.54) as secondary, lower-priority threats. The 22-zone segmentation distinguishes deep-water cores (critical) from shallower water margins (high), enabling tiered prioritization. SAR-based mapping carries inherent limitations: smooth non-water surfaces (wet pavement, certain roofs, calm reservoirs) can produce false positives, and dense urban geometry may cause layover/shadow gaps. Affected-area and damage-percent figures should be treated as model estimates pending ground truth. The vulnerability score (8.2) integrates population exposure (540,000), hospital exposure (14), road denial (89 km), and school disruption (67) into a single prioritization index.

Recommendations

- Immediately verify operational status and flood defenses at Lady Reading Hospital (HIGH risk) and pre-position medical evacuation assets.
- Activate Evacuation Route 1 (71.49,33.97 -> 71.53,34.00 -> 71.59,34.06) toward higher ground northeast, and confirm it remains above floodwater before mass movement.
- Conduct rapid ground/drone verification of critical zones FZ-01 and FZ-02 to validate SAR-derived water extents.
- Restore or bypass the 89 km of blocked roadway with priority on corridors linking the 14 at-risk hospitals.
- Establish shelter and triage capacity sized for a population-affected estimate of up to 540,000.
- Issue targeted warnings to the 67 affected schools and confirm full evacuation/closure status.
- Deploy water rescue teams to deep-water class areas first, then high-severity water margins.

Response Priorities

- Life safety: evacuate and rescue populations in critical deep-water zone FZ-01 and high-severity FZ-02.
- Protect and sustain healthcare: secure 14 at-risk hospitals, prioritizing Lady Reading Hospital.
- Restore mobility: clear or bypass 89 km of blocked roads to enable rescue and resupply.
- Validate intelligence: confirm SAR flood extents and damage estimates with ground truth.
- Shelter and welfare: stand up capacity for displaced population near affected districts.
- Monitor secondary hazards: maintain watch on MEDIUM earthquake risk.

Assumptions

- Population-affected figure of 540,000 reflects all persons within or adjacent to the mapped flood footprint, not confirmed displacement.
- Damage percent (24.3%) and affected area (153.37 km²) are model-derived from SAR and not yet field-verified.
- Evacuation Route 1 is assumed passable and above current floodwater at time of activation.
- Critical facility risk ratings reflect proximity-based exposure modeling.
- Reported flood extent corresponds to conditions at the time of the Sentinel-1 acquisition.

Limitations

- SAR VV/VH water detection can misclassify smooth non-water surfaces, producing false positives or negatives.
- 42% cloud cover precluded optical validation, removing an independent cross-check.
- Urban layover, shadow, and double-bounce effects may leave detection gaps in dense built-up areas.
- Only two named critical facilities were provided; the full set of 14 at-risk hospitals lacks individual geolocation here.
- Single-scene snapshot does not capture floodwater rise/recession dynamics between acquisitions.
- Confidence scores reflect model certainty and do not replace on-ground confirmation.

Model Source Note

Featherless detailed generation failed; AI/ML fallback was used. Executive summary generated using AI/ML API.

Detailed Report	aiml_fallback
Executive Summary	aiml
Fallback Used	True
Featherless Model	moonshotai/Kimi-K2.6

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless Kimi K2.6 unavailable, AI/ML fallback used
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally